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LIQUID CIRCULATION DEVICE, LIQUID DISCHARGE APPARATUS, AND IMAGE FORMING APPARATUS

Abstract

A liquid circulation device includes a supply tank to store a liquid and a gas in a first space, a collection tank to store the liquid and the gas in a second space, a circulation path, a liquid path, a liquid feed pump, a first gas path communicating with the first space, a first gas pump to supply gas, a second gas path communicating with the second space, a second gas pump to vacuum the gas, a gas bypass connecting the first space and the second space, and circuitry to control the liquid feed pump to flow the liquid through the liquid path at a first flow rate and control the first gas pump and the second gas pump to flow the gas from the first space to the second space through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2024-030310, filed on Feb. 29, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a liquid circulation device, a liquid discharge apparatus, and an image forming apparatus.

Related Art

[0003] A liquid discharge apparatus including a circulation-type head as a liquid discharge head is known in the art.

SUMMARY

[0004] The present disclosure described herein provides an improved liquid circulation device including a supply tank to store a liquid to be supplied to a head and a gas in a first space above a surface of the liquid in the supply tank, a collection tank to store the liquid collected from the head and the gas in a second space above a surface of the liquid in the collection tank, a circulation path including a first circulation path connecting the supply tank and an inlet of the head and a second circulation path connecting an outlet of the head and the collection tank to circulate the liquid through the head, a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas path communicating with the first space in the supply tank, a first gas pump to supply gas to the first space through the first gas path, a second gas path communicating with the second space in the collection tank, a second gas pump to vacuum the gas from the collection tank through the second gas path, a gas bypass connecting the first space in the supply tank and the second space in the collection tank, and circuitry to control the liquid feed pump to flow the liquid through the liquid path at a first flow rate and control the first gas pump and the second gas pump to flow the gas from the first space to the second space through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

[0005] Further, the present disclosure described herein provides an improved liquid discharge apparatus including a head to discharge a liquid, a supply tank to store the liquid and a gas, a collection tank to store the liquid and the gas, a circulation path including a first circulation path connecting the supply tank and the head and a second circulation path connecting the head and the collection tank to circulate the liquid, a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas pump to supply gas to the supply tank, a second gas pump to vacuum the gas from the collection tank, and a gas bypass connecting the supply tank and the collection tank.

[0006] Further, the present disclosure described herein provides an improved image forming apparatus including a head to discharge a liquid onto a medium, a supply tank to store the liquid and a gas, a collection tank to store the liquid and the gas, a circulation path including a first circulation path connecting the supply tank and the head and a second circulation path connecting the head and the collection tank to circulate the liquid, a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas pump to supply gas to the supply tank, a second gas pump to vacuum the gas from the collection tank, a gas bypass connecting the supply tank and the collection tank, and circuitry controls the liquid feed pump to flow the liquid through the liquid path at a first flow rate and controls the first gas pump and the second gas pump to flow the gas from the supply

tank to the collection tank through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0008] FIG. 1 is a schematic view of a printer as a liquid discharge apparatus;

[0009] FIG. 2 is a plan view of a head unit of the printer of FIG. 1;

[0010] FIG. 3 is an external perspective view of a liquid discharge head;

[0011] FIG. 4 is a cross-sectional view of a liquid discharge head taken in the direction (i.e., a longitudinal direction of a liquid chamber) orthogonal to a nozzle array direction of the liquid discharge head;

[0012] FIG. 5 is a diagram illustrating the piping of a liquid circulation device;

[0013] FIG. 6 is a functional block diagram of a controller of a liquid discharge apparatus including a liquid circulation device;

[0014] FIG. 7 is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a comparative example;

[0015] FIGS. 8A and 8B are schematic diagrams each illustrating a configuration of a part of a liquid circulation device according to an embodiment of the present disclosure;

[0016] FIG. 9 is a schematic diagram illustrating another configuration of a part of a liquid circulation device;

[0017] FIG. 10 is a schematic diagram illustrating yet another configuration of a part of a liquid circulation device;

[0018] FIG. 11 is a schematic diagram illustrating still another configuration of a part of a liquid circulation device; and

[0019] FIG. 12 is a schematic diagram illustrating still yet another configuration of a part of a liquid circulation device.

[0020] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0021] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0022] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0023] A liquid circulation device, a liquid discharge apparatus, and an image forming apparatus according to embodiments of the present disclosure are described below with reference to the drawings. Embodiments of the present disclosure are not limited to the embodiments described below and may be other embodiments than the embodiments described below. The following embodiments may be modified by, for example, addition, modification, or omission within the scope that would be obvious to one skilled in the art. Any aspects having advantages as described

for the following embodiments according to the present disclosure are included within the scope of the present disclosure.

[0024] A printer as a liquid discharge apparatus will be described below with reference to FIGS. 1 to 2. The printer and the liquid discharge apparatus include a liquid circulation device which will be described later.

[0025] FIG. 1 is a schematic view of the printer. FIG. 2 is a plan view of a head unit of the printer of FIG. 1.

[0026] A printer **1000** as the liquid discharge apparatus includes a feeder **1** to feed a continuous medium **10**, a guide conveyor **3** as a conveyor to guide and convey the continuous medium **10**, fed from the feeder **1**, to a printing device **5**, the printing device **5** to discharge a liquid onto the continuous medium **10** to form an image on the continuous medium **10**, a dryer **7** to dry the continuous medium **10**, and a carrier **9** to feeds the dried continuous medium **10** outward. The continuous medium **10** is a discharge target or a medium onto which the discharged liquid is applied.

[0027] The continuous medium **10** is fed from a winding roller **11** of the feeder **1**, guided and conveyed with rollers of the feeder **1**, the guide conveyor **3**, the dryer **7**, and the carrier **9**, and wound around a take-up roller **91** of the carrier **9**.

[0028] In the printing device **5**, the continuous medium **10** is conveyed on a conveyance guide **59** so as to face a head unit **50** and a head unit **55**. An image is formed with liquid discharged from the head unit **50**, and post-treatment is performed with treatment liquid discharged from the head unit **55**.

[0029] The head unit **50** includes, for example, full-line head arrays **51K**, **51C**, **51M**, and **51Y** for four colors from the upstream side in a conveyance direction of the continuous medium **10**. The full-line head arrays **51K**, **51C**, **51M**, and **51Y** may be referred to simply as the “head array **51**” when colors are not distinguished.

[0030] Each of the head arrays **51** is a liquid discharge device to discharge liquid (e.g., ink) of black (K), cyan (C), magenta (M), or yellow (Y) onto the continuous medium **10** conveyed in the conveyance direction. The number and types of colors are not limited to the above-described four colors of K, C, M, and Y and may be any other suitable number and types.

[0031] In each head array **51**, for example, as illustrated in FIG. 2, liquid discharge heads **100** are disposed in a staggered arrangement on a base **52** to form the head array **51**. The head array **51** is not limited to such an arrangement. The liquid discharge head **100** may be referred to simply as a head **100** in the following description.

[0032] An example of the liquid discharge head is described below with reference to FIGS. 3 and 4. In the present embodiment, the liquid discharge head is a circulation-type liquid discharge head.

[0033] FIG. 3 is an external perspective view of the liquid discharge head. FIG. 4 is a cross-sectional view of the liquid discharge head of FIG. 3 in a direction (i.e., the longitudinal direction of a liquid chamber) orthogonal to a nozzle array direction.

[0034] The liquid discharge head **100** includes a nozzle plate **101**, a channel plate **102**, and a diaphragm substrate **103** laminated one on another and bonded to each other. The diaphragm substrate **103** serves as a wall of a liquid chamber defined by the channel plate **102**. The liquid discharge head **100** further includes a piezoelectric actuator **111** to displace a vibration portion (diaphragm) **130** of the diaphragm substrate **103**, a common-chamber substrate **120** that also serves as a frame of the liquid discharge head **100**, and a cover **129**. A portion of the head **100** constructed by the channel plate **102** and the diaphragm substrate **103** is referred to as a channel substrate **140**.

[0035] The nozzle plate **101** has multiple nozzles **104**, which are arranged in the nozzle array direction, to discharge a liquid.

[0036] The channel plate **102** has through holes and groves that define individual liquid chambers **106**, supply-side liquid restrictors **107**, and liquid inlets **108**. The individual liquid chambers **106** communicate with the nozzles **104** via nozzle communication channels **105**, respectively. The

supply-side liquid restrictors **107** communicate with the individual liquid chambers **106**, respectively. The liquid inlets **108** communicate with the supply-side liquid restrictors **107**, respectively. The nozzle communication channels **105** connect the nozzles **104** and the individual liquid chambers **106**, respectively. The liquid inlets **108** communicate with a supply-side common liquid chamber **110** through openings **109** of the diaphragm substrate **103**.

[0037] The diaphragm substrate **103** includes the deformable vibration portion **130** serving as the wall of the individual liquid chambers **106** of the channel plate **102**. In the present embodiment, the diaphragm substrate **103** has, but is not limited to, a double-layer structure and includes a first layer forming a thin portion and a second layer forming a thick portion in this order from the channel plate **102**. A portion of the first layer corresponding to the individual liquid chamber **106** forms the deformable vibration portion **130**.

[0038] The piezoelectric actuator **111** is disposed on one side of the diaphragm substrate **103** opposite the other side facing the individual liquid chamber **106**. The piezoelectric actuator **111** includes an electromechanical transducer as a driving device (actuator device or pressure generator) to deform the vibration portion **130** of the diaphragm substrate **103**.

[0039] The piezoelectric actuator **111** includes a piezoelectric element **112** bonded on a base **113**. The piezoelectric element **112** is grooved by half-cut dicing to form a comb shape including a desired number of pillar-shaped elements that are arranged at certain intervals.

[0040] The piezoelectric element **112** is bonded to a projection **130a**, which is an island-shaped thick portion on the vibration portion **130** of the diaphragm substrate **103**. A flexible wiring board **115** is connected to the piezoelectric element **112**.

[0041] The common-chamber substrate **120** defines the supply-side common liquid chamber **110** and a delivery-side common liquid chamber **150**. The supply-side common liquid chamber **110** communicates with a supply port **171** (i.e., an inlet of the liquid discharge head **100**), and the delivery-side common liquid chamber **150** communicates with a delivery port **181** (i.e., an outlet of the liquid discharge head **100**).

[0042] The common-chamber substrate **120** includes a first common-chamber substrate **121** and a second common-chamber substrate **122**. The first common-chamber substrate **121** is bonded to the diaphragm substrate **103** of the channel substrate **140**. The second common-chamber substrate **122** is laminated on and bonded to the first common-chamber substrate **121**.

[0043] The first common-chamber substrate **121** defines a downstream common chamber **110A** and the delivery-side common liquid chamber **150**. The downstream common chamber **110A** is a part of the supply-side common liquid chamber **110** communicating with the liquid inlets **108**. The delivery-side common liquid chamber **150** communicates with delivery channels **151**. The second common-chamber substrate **122** defines an upstream common chamber **110B** which is the other part of the supply-side common liquid chamber **110**.

[0044] The channel plate **102** further defines the delivery channels **151** extending in the in-plane direction of the channel plate **102**. The delivery channels **151** communicate with the individual liquid chambers **106** via the nozzle communication channels **105**, respectively. The delivery channels **151** also communicate with the delivery-side common liquid chamber **150**.

[0045] In the liquid discharge head **100**, for example, the voltage to be applied to the piezoelectric element **112** is lowered from a reference potential (intermediate potential) so that the piezoelectric element **112** contracts to pull the vibration portion **130** of the diaphragm substrate **103** to increase the volume of the individual liquid chamber **106**. As a result, liquid flows into the individual liquid chamber **106**.

[0046] Then, the voltage to be applied to the piezoelectric element **112** is increased to expand the piezoelectric element **112** in the direction of lamination, and the vibration portion **130** of the diaphragm substrate **103** is deformed in a direction toward the nozzle **104** to reduce the volume of the individual liquid chamber **106**. As a result, the liquid in the individual liquid chamber **106** is pressurized and discharged from the nozzle **104**.

[0047] The liquid that has not been discharged from the nozzle **104** passes through the nozzle **104** and is delivered to the delivery-side common liquid chamber **150** through the delivery channel **151**. Then, the liquid is delivered from the delivery-side common liquid chamber **150** to an external liquid circulation path (e.g., a liquid circulation device) and supplied to the supply-side common liquid chamber **110** again through the external liquid circulation path.

[0048] The method of driving the liquid discharge head is not limited to the above-described example (pull-push discharge). For example, pull discharge or push discharge may be performed in accordance with the way to apply a drive waveform.

[0049] The circulation-type liquid discharge head connected to a liquid circulation device of the present embodiment may be an individual liquid chamber type head in which liquid circulates in the vicinity of nozzles, or may be a head in which liquid circulates only in a common liquid chamber.

[0050] A liquid circulation device included in the liquid discharge apparatus and the printer according to the present embodiment is described below with reference to FIG. 5. FIG. 5 is a diagram illustrating the piping of the liquid circulation device. Arrows in FIG. 5 indicate the direction of flow of liquid.

[0051] The liquid circulation device of the present embodiment is connected to multiple heads **100** arranged in a line. Liquid can be circulated through the multiple heads **100**. In FIG. 5, only the heads **100** at both ends in the head array **51** are illustrated.

[0052] The arrangement direction of the multiple heads **100** is preferably the same as the width direction of the discharge target (e.g., the continuous medium **10** in FIG. 1) to which the discharged liquid is applied.

[0053] A liquid circulation device **200** includes a main tank (liquid tank) **201**, a supply tank **210**, a collection tank **220**, and a sub tank **290**. The main tank **201** is a liquid storage unit that stores liquid to be discharged from the head **100**. The liquid circulation device **200** further includes a first liquid feed pump **202A** and a second liquid feed pump **202B**, which are disposed in a liquid path, and a third liquid feed pump **209**.

[0054] The liquid circulation device **200** further includes a first manifold **230** and a second manifold **240** communicating with a first head tank (pressurization head tank) **300a** and a second head tank (depressurization head tank) **300b** of each of multiple heads **100**, and a deaerator **260** which is a degassing unit for removing dissolved gas in the liquid. The first head tank **300a** and the second head tank **300b** may be referred to as head tank **300** when not distinguished in the following description.

[0055] The liquid stored in the main tank **201** is fed (supplied) to the sub tank **290** by the third liquid feed pump **209** via a liquid feed path **289** including a filter **205**. The sub tank **290** includes a liquid level detector **291** and a solenoid valve **292** constructing an air release mechanism for releasing air inside the sub tank **290** to the atmosphere.

[0056] The sub tank **290** and the supply tank **210** are connected to each other through a first liquid path **203A**, and the first liquid feed pump **202A** is disposed in the first liquid path **203A**. In addition, the sub tank **290** and the supply tank **210** are connected to each other through a return liquid path **285**, and a solenoid valve **287** is disposed in the return liquid path **285**.

[0057] The supply tank **210** has a gas space **210a** above the surface of liquid. In other words, the supply tank **210** stores both the liquid and the air. The supply tank **210** includes a liquid level detector **211** and a solenoid valve **212** constructing an air release mechanism for releasing air inside the supply tank **210** to the atmosphere.

[0058] The sub tank **290** and the collection tank **220** are connected to each other through a second liquid path **203B**, and the second liquid feed pump **202B** is disposed in the second liquid path **203B**. In addition, the sub tank **290** and the collection tank **220** are connected to each other through a return liquid path **286**, and a solenoid valve **288** is disposed in the return liquid path **286**.

[0059] The collection tank **220** has a gas space **220a** above the surface of liquid. In other words, the

collection tank **220** stores both the liquid and the air. The collection tank **220** includes a liquid level detector **221** and a solenoid valve **222** constructing an air release mechanism for releasing air inside the collection tank **220** to the atmosphere.

[0060] The supply tank **210** is connected to the first manifold **230** through a liquid path **281** including the deaerator **260** and a filter **261**. The first manifold **230** communicates with the supply port **171** of the head **100** illustrated in FIG. 3 via a supply path **231**. The supply path **231** is connected to the supply port **171** of the head **100** via the first head tank **300a**. The supply path **231** is provided with a solenoid valve **232** for opening and closing the path on the upstream side of the first head tank **300a**. The solenoid valves **232** are provided for the heads **100**, respectively, and can be individually controlled to be opened and closed. In other words, the number of the solenoid valves **232** is equal to the number of the heads **100**. The first manifold **230** is provided with a pressure sensor **233**.

[0061] The collection tank **220** is connected to the second manifold **240** via a liquid path **282**. The second manifold **240** communicates with the delivery port **181** of the head **100** illustrated in FIG. 3 via a delivery path **241**. The delivery path **241** is connected to the delivery port **181** of the head **100** via the second head tank **300b**. The delivery path **241** is provided with a solenoid valve **242** for opening and closing the path on the downstream side of the second head tank **300b**. The solenoid valves **242** are provided for the heads **100**, respectively, and can be individually controlled to be opened and closed. In other words, the number of the solenoid valves **242** is equal to the number of the heads **100**. The second manifold **240** is provided with a pressure sensor **243**.

[0062] A manifold connection path **270** connects the first manifold **230** and the second manifold **240**. The manifold connection path **270** is provided with a solenoid valve **271** near the first manifold **230** and a solenoid valve **272** near the second manifold **240**.

[0063] Thus, a circulation path is formed by a path from the sub tank **290**, via the first liquid path **203A**, the supply tank **210**, the liquid path **281**, the deaerator **260**, the first manifold **230**, the head **100**, the second manifold **240**, the liquid path **282**, the collection tank **220**, and the second liquid path **203B**, to the sub tank **290**. In other words, the circulation path includes a supply-side circulation path (i.e., a first circulation path) connecting the supply tank **210** and the inlet (i.e., the supply port **171**) of the head **100** and a collection-side circulation path (i.e., a second circulation path) connecting the outlet (i.e., the delivery port **181**) of the head **100** and the collection tank **220**.

[0064] The liquid path **203** (i.e., the first liquid path **203A** and the second liquid path **203B**) connects the supply tank **210** and the collection tank **220** via the sub tank **290**. In the following description regarding the feeding of the liquid between the supply tank **210** and the collection tank **220**, the sub tank **290** may be omitted.

[0065] The solenoid valves **232**, **242**, **271**, and **272** are opened and closed to cut off the head **100** from the circulation path. The solenoid valves **232**, **242**, **271**, and **272** switch between a first path and a second path. The first path includes the manifold connection path **270** as a part of the circulation path. The second path includes the head **100** as a part of the circulation path and cuts off the manifold connection path **270** from the circulation path.

[0066] When the solenoid valves **232** and **242** are closed and the solenoid valves **271** and **272** are opened, the circulation path includes the manifold connection path **270** and does not include the head **100** to form the first path. On the other hand, when the solenoid valves **232** and **242** are opened and the solenoid valves **271** and **272** are closed, the circulation path includes the head **100** and does not include the manifold connection path **270** to form the second path.

[0067] The supply tank **210**, the collection tank **220**, the first liquid feed pump **202A**, and the second liquid feed pump **202B** construct a unit for generating pressures for circulating the liquid in the circulating path.

[0068] The supply and circulation of the liquid will be described below. Liquid is fed from the main tank **201** to the sub tank **290** as follows. When the liquid level detector **291** detects a shortage of the liquid in the sub tank **290**, the third liquid feed pump **209** supplies the liquid from the main

tank **201** to the sub tank **290** via the liquid feed path **289** until the liquid level detector **291** detects that the liquid level is full.

[0069] Liquid is fed from the sub tank **290** to the supply tank **210** as follows. The first liquid feed pump **202A** supplies the liquid from the sub tank **290** to the supply tank **210** via the first liquid path **203A**.

[0070] Liquid is fed from the collection tank **220** to the sub tank **290** as follows. The second liquid feed pump **202B** supplies the liquid from the collection tank **220** to the sub tank **290** via the second liquid path **203B**.

[0071] Liquid is fed from the supply tank **210** to the head **100** and from the head **100** to the collection tank **220** as follows. The first liquid feed pump **202A** supplies the liquid to the supply tank **210** until the pressure sensor **233** indicates a target pressure (for example, a positive pressure), and the second liquid feed pump **202B** supplies the liquid to the sub tank **290** until the pressure sensor **243** indicates a target pressure (for example, a negative pressure). As a result, a pressure difference is generated between the supply tank **210** and the collection tank **220**. In response to this pressure difference, the liquid can be circulated from the supply tank **210**, via the liquid path **281**, the filter **261**, the deaerator **260**, the first manifold **230**, multiple supply paths **231**, multiple first head tanks **300a**, multiple heads **100**, multiple delivery paths **241**, multiple second head tanks **300b**, the second manifold **240**, and the liquid path **282**, to the collection tank **220**. At this time, the solenoid valves **232** and **242** are opened and the solenoid valves **271** and **272** are closed.

[0072] On the other hand, when the first liquid feed pump **202A** and the second liquid feed pump **202B** are driven with the solenoid valves **232** and **242** closed and the solenoid valves **271** and **272** opened to generate a pressure difference, the liquid can be circulated from the supply tank **210**, via the liquid path **281**, the filter **261**, the deaerator **260**, the first manifold **230**, the manifold connection path **270**, the second manifold **240**, and the liquid path **282**, to the collection tank **220** in response to the pressure difference.

[0073] The liquid level detector **211**, **221**, and **291** in the respective tanks are not limited to any particular detector, and can employ, for example, a float type detection method, a detection method using a two or more electrode pins for detecting a voltage, or a liquid level detection method using a laser to detect the presence or absence of the liquid.

[0074] The supply tank **210**, the collection tank **220**, and the sub tank **290** are provided with the solenoid valves **212**, **222**, and **292**, respectively, as air release mechanisms. The solenoid valve enables air inside the tank to communicate with the atmosphere.

[0075] The liquid circulation device **200** includes an air supply-and-exhaust system that adjusts the internal pressure of the supply tank **210** and the collection tank **220**. The air supply-and-exhaust system includes a gas storage unit (supply-side air tank) connected to the supply tank **210** and a gas storage unit (collection-side air tank) connected to the collection tank **220**.

[0076] The air supply-and-exhaust system further includes a supply-side air path **83** (i.e., a first gas path) that connects the supply tank **210** and the supply-side air tank, a supply-side air pump **81** (i.e., a first gas pump) disposed in the supply-side air path **83** to pressurize (or depressurize in some cases) the supply tank **210**, a collection-side air path **84** (i.e., a second gas path) that connects the collection tank **220** and the collection-side air tank, and a collection-side air pump **82** (i.e., a second gas pump) disposed in the collection-side air path **84** to depressurize the collection tank **220**. The supply-side air pump **81** may depressurize the supply tank **210** to a negative pressure lower than that of the collection tank **220**.

[0077] The functions of the gas space **210a** of the supply tank **210** and the gas space **220a** of the collection tank **220** will be described below. In the gas space **210a** and the gas space **220a**, the surface of the liquid is in contact with a gas (e.g., air or inert gas).

[0078] When a pressurized state (positive pressure) of the gas is generated in the supply tank **210** and a depressurized state (negative pressure) of the gas is generated in the collection tank **220**, it can be said the supply tank **210** and the collection tank **220** store the positive pressure and the

negative pressure, respectively, due to the compressibility of the gas. In this case, the gas is considered to be similar to a capacitor component when being expressed by an equivalent electric circuit, and can also be expressed as compliance (elastic component).

[0079] When the liquid feed pumps **202** (i.e., the first liquid feed pump **202A** and the second liquid feed pump **202B**), which are disposed in the liquid paths **203** (i.e., the first liquid path **203A** and the second liquid path **203B**) communicating with the supply tank **210** and the collection tank **220**, are driven, a pressure change (pulsation) is generated. When the pressure change propagates to the meniscus of the liquid in the nozzle **104** through the liquid path **203**, the liquid may overflow or may entrain bubbles. The compliance (elastic component) can reduce such a pressure change. A typical gas has compressibility and can serve as the compliance (elastic component). Accordingly, the gas space **210a** and the gas space **220a** can reduce a large pressure change.

[0080] Further, the liquid circulation device **200** includes a bypass **90** (i.e., a gas bypass) communicating with the gas space **210a** and the gas space **220a**. The bypass **90** can reduce the pulsation of the liquid generated when the liquid is fed from the collection tank **220** to the supply tank **210**.

[0081] A controller of the liquid discharge apparatus (the printer **1000**) is described below with reference to FIG. **6**. FIG. **6** is a functional block diagram of the controller.

[0082] In FIG. **6**, a controller **500** as circuitry includes a main controller **500A** that includes a central processing unit (CPU) **501**, a read-only memory (ROM) **502**, and a random-access memory (RAM) **503**. The CPU **501** administrates the control of the entire liquid discharge apparatus. The ROM **502** stores fixed data, such as various programs including programs executed by the CPU **501**. The RAM **503** temporarily stores image data and other data.

[0083] The controller **500** further includes a nonvolatile RAM (NVRAM) **504** that is rewritable and retains data while the apparatus is powered off. The controller **500** further includes an application-specific integrated circuit (ASIC) **505** that performs image processing, such as various signal processing and sorting on image data, and processing of input and output signals for other control. The controller **500** transmits and receives data to and from a printer driver **590** via a host interface (I/F) **506**.

[0084] The controller **500** further includes a print controller **508** including a data transfer unit, a drive signal generation unit, and a bias voltage output unit for controlling a head driver **509** to drive each head **100** of the head unit **50**. The head driver **509** is a drive integrated circuit (IC) for driving each head **100**.

[0085] The controller **500** further includes a solenoid valve controller **510** that controls the driving of the solenoid valves **232**, **242**, **271**, and **272** and the solenoid valves **212**, **222**, **292**, **287**, and **288** in a group of solenoid valves **550**.

[0086] The controller **500** further includes a supply system controller **511** that controls the driving of the third liquid feed pump **209**. The controller **500** further includes a pressure system controller **512** that controls the driving of the first liquid feed pump **202A**, the second liquid feed pump **202B**, the supply-side air pump **81**, and the collection-side air pump **82** to control (change) flow rates of liquid and gas. The first liquid feed pump **202A** and the second liquid feed pump **202B** may be referred to collectively as a liquid feed pump **202** unless distinguished from each other in the following description.

[0087] The controller **500** further includes an input/output (I/O) unit **513**. The I/O unit **513** can perform the processing of data from sensors to acquire data from various sensors **515** and detection results by the pressure sensors **233** and **243**. The I/O unit **513** extracts data for controlling the liquid discharge apparatus, and the print controller **508**, the solenoid valve controller **510**, the supply system controller **511**, and the pressure system controller **512** use the data to control the respective components. The controller **500** is connected to a control panel **514** to input and display data for the liquid discharge apparatus.

Configuration of Part of Liquid Circulation Device

[0088] FIG. 7 is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a comparative example. In the liquid circulation device of FIG. 7, the supply tank **210**, the collection tank **220**, and the head **100** form a liquid circulation system to discharge liquid from the head **100** while circulating the liquid. In the liquid circulation system, the liquid is fed to the head **100** by a pressure difference between the supply tank **210** and the collection tank **220**. The liquid feed pump **202** is disposed in the liquid path **203** to feed the liquid from the collection tank **220** to the supply tank **210**.

[0089] When the liquid is supplied toward the head **100** and the liquid in the supply tank **210** decreases, the volume of the gas space **210a** of the supply tank **210** increases and the internal pressure of the supply tank **210** decreases. The supply-side air pump **81** is operated to supply gas to the supply tank **210** to adjust the internal pressure of the supply tank **210**.

[0090] On the other hand, when the liquid is collected into the collection tank **220** and the liquid in the collection tank **220** increases, the volume of the gas space **220a** of the collection tank **220** decreases and the internal pressure of the collection tank **220** increases. The collection-side air pump **82** is operated to suck (vacuum) the gas in the collection tank **220** to adjust the internal pressure of the collection tank **220**.

[0091] When the liquid in the supply tank **210** decreases and the liquid in the collection tank **220** increases, the liquid feed pump **202** feeds the liquid from the collection tank **220** to the supply tank **210**.

[0092] When the liquid is supplied to the supply tank **210** and the liquid in the supply tank **210** increases, the volume of the gas space **210a** of the supply tank **210** decreases and the internal pressure of the supply tank **210** increases. On the other hand, when the liquid in the collection tank **220** decreases, the volume of the gas space **220a** of the collection tank **220** increases and the internal pressure of the collection tank **220** decreases.

[0093] At this time, even if the supply-side air pump **81** is operated, the internal pressure of the supply tank **210** does not decrease, and the pulsation of the pressure may be generated until the internal pressure decreases due to the decrease in the amount of liquid. Similarly, the internal pressure of the collection tank **220** does not increase even if the collection-side air pump **82** is operated, and the pulsation of the pressure may be generated until the internal pressure increases due to the increase in the amount of liquid.

[0094] In the present embodiment, the liquid circulation device **200** includes the bypass **90** that prevents the pulsation of the pressure described above. A description is given below of a liquid circulation device according to an embodiment of the present disclosure.

First Embodiment

[0095] FIGS. **8A** and **8B** are schematic diagrams each illustrating a configuration of a part of a liquid circulation device according to a first embodiment of the present disclosure. Arrows in FIGS. **8A** and **8B** indicate the direction of flow of liquid and gas.

[0096] The liquid circulation device of the present embodiment includes the supply tank **210** and the collection tank **220** to store a liquid, the head **100** to discharge the liquid, the path to circulate the liquid via the head **100**, the liquid feed pump **202** disposed in the liquid path **203** to feed the liquid from the collection tank **220** to the supply tank **210**, the supply-side air pump **81** disposed in the supply-side air path **83** communicating with the gas space **210a** of the supply tank **210**, the collection-side air pump **82** disposed in the collection-side air path **84** communicating with the gas space **220a** of the collection tank **220**, and the bypass **90** connecting the gas space **210a** of the supply tank **210** and the gas space **220a** of the collection tank **220** and serving as the flow path for gas flowing from the supply tank **210** to the collection tank **220**. The flow rate of the gas flowing through the bypass **90** is smaller than the maximum flow rate of the liquid flowing through the liquid feed pump **202**.

[0097] As illustrated in FIG. **8B**, when a second flow rate **F2** represents the flow rate of the gas flowing through the bypass **90** and a first flow rate **F1** represents the flow rate of the liquid flowing

through the liquid feed pump **202**, the liquid circulation device of the present embodiment satisfies the relationship of $F1.sub.MAX > F2$.

[0098] The “flow rate” is the amount of movement of liquid or gas per unit time, and the “maximum flow rate” is the maximum flow rate of liquid or gas at which the pump can supply the liquid or the gas.

[0099] The bypass **90** allows gas to flow from the supply tank **210** to the collection tank **220**. Thus, even when the liquid feed pump **202** is operated and the liquid in the supply tank **210** increases, the gas in the gas space **210a** is exhausted through the bypass **90**. Accordingly, the increase in the internal pressure of the supply tank **210** is prevented, and the pulsation of the pressure is reduced.

[0100] Further, even when the liquid feed pump **202** is operated and the liquid in the collection tank **220** decreases, the gas is supplied to the gas space **220a** through the bypass **90**. Accordingly, the decrease in the internal pressure of the collection tank **220** is prevented, and the pulsation of the pressure is reduced.

[0101] On the other hand, when the liquid feed pump **202** is not operated, the feeding of the liquid is stopped and the internal pressure of the supply tank **210** decreases, but the supply-side air pump **81** is operated to supply the gas to the supply tank **210** to adjust the internal pressure of the supply tank **210**. Further, although the internal pressure of the collection tank **220** increases, the collection-side air pump **82** is operated to suck (vacuum) the gas in the collection tank **220** to adjust the internal pressure of the collection tank **22**.

[0102] Since the second flow rate $F2$ of the gas flowing through the bypass **90** is smaller than the maximum value of the first flow rate $F1$ of the liquid flowing through the liquid feed pump **202**, the fluctuation of the internal pressure of the supply tank **210** and the collection tank **220** does not exceed the range of the fluctuation of the internal pressure due to the operation of the liquid feed pump **202**.

[0103] With the configuration of the present embodiment, the internal pressure of the supply tank **210** and the collection tank **220** can be easily adjusted within a certain range, and the pulsation of the pressure can be reduced with a simple configuration.

Flow Rate of Supply-Side Air Pump

[0104] In the liquid circulation device of the present embodiment, the maximum flow rate of the gas flowing through the supply-side air pump **81** is preferably larger than the flow rate of the gas flowing through the bypass **90**.

[0105] As illustrated in FIG. **8B**, when the second flow rate $F2$ represents the flow rate of the gas flowing through the bypass **90** and a third flow rate $F3$ represents the flow rate of the gas flowing through the supply-side air pump **81**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F3.sub.MAX > F2$.

[0106] With such a configuration, the fluctuation of the internal pressure of the supply tank **210** due to the gas flowing out through the bypass **90** does not exceed the range of the fluctuation of the internal pressure due to the operation of the supply-side air pump **81**. Accordingly, the internal pressure of the supply tank **210** can be easily adjusted within a certain range.

[0107] The flow rate of the gas flowing through the supply-side air pump **81** is preferably larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid supplied from the supply tank **210** to the head **100**.

[0108] As illustrated in FIG. **8B**, when the third flow rate $F3$ represents the flow rate of the gas flowing through the supply-side air pump **81**, the second flow rate $F2$ represents the flow rate of the gas flowing through the bypass **90**, and a fourth flow rate $F4$ represents the flow rate of the liquid supplied from the supply tank **210** to the head **100**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F3 > (F2 + F4)$.

[0109] With such a configuration, the internal pressure of the supply tank **210** can be easily adjusted within a certain range in consideration of the fluctuation of the internal pressure (the decrease in the internal pressure) of the supply tank **210** due to the circulation of the liquid.

[0110] The flow rate of the gas flowing through the supply-side air pump **81** is preferably larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid supplied from the supply tank **210** to the head **100**, which is increased by the amount of the liquid that has been discharged from the head **100**.

[0111] As illustrated in FIG. **8B**, when the third flow rate **F3** represents the flow rate of the gas flowing through the supply-side air pump **81**, the second flow rate **F2** represents the flow rate of the gas flowing through the bypass **90**, the fourth flow rate **F4** represents the flow rate of the liquid supplied from the supply tank **210** to the head **100**, and a seventh flow rate **F7** represents the flow rate of the liquid that has been discharged from the head **100**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F3 > (F2 + F4 + F7)$.

[0112] With such a configuration, the internal pressure of the supply tank **210** can be reliably adjusted within a certain range in consideration of the fluctuation of the internal pressure (the decrease in the internal pressure) of the supply tank **210** due to the amount of the liquid discharged from the head **100**.

Flow Rate of Collection-Side Air Pump

[0113] In the liquid circulation device of the present embodiment, the maximum flow rate of the gas flowing through the collection-side air pump **82** is preferably larger than the flow rate of the gas flowing through the bypass **90**.

[0114] As illustrated in FIG. **8B**, when the second flow rate **F2** represents the flow rate of the gas flowing through the bypass **90** and a fifth flow rate **F5** represents the flow rate of the gas flowing through the collection-side air pump **82**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F5_{\text{sub.MAX}} > F2$.

[0115] With such a configuration, the fluctuation of the internal pressure of the collection tank **220** due to the gas flowing into the collection tank **220** through the bypass **90** does not exceed the range of the fluctuation of the internal pressure due to the operation of the collection-side air pump **82**. Accordingly, the internal pressure of the collection tank **220** can be easily adjusted within a certain range.

[0116] The flow rate of the gas flowing through the collection-side air pump **82** is preferably larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid collected into the collection tank **220** via the head **100**.

[0117] As illustrated in FIG. **8B**, when the fifth flow rate **F5** represents the flow rate of the gas flowing through the collection-side air pump **82**, the second flow rate **F2** represents the flow rate of the gas flowing through the bypass **90**, and a sixth flow rate **F6** represents the flow rate of the liquid collected into the collection tank **220**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F5 > (F2 + F6)$.

[0118] With such a configuration, the internal pressure of the collection tank **220** can be easily adjusted within a certain range in consideration of the fluctuation of the internal pressure (the increase in the internal pressure) of the collection tank **220** due to the circulation of the liquid.

[0119] The flow rate of the gas flowing through the collection-side air pump **82** is preferably larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid collected into the collection tank **220** via the head **100**, which is decreased by the amount of the liquid that has been discharged from the head **100**.

[0120] As illustrated in FIG. **8B**, when the fifth flow rate **F5** represents the flow rate of the gas flowing through the collection-side air pump **82**, the second flow rate **F2** represents the flow rate of the gas flowing through the bypass **90**, the sixth flow rate **F6** represents the flow rate of the liquid collected into the collection tank **220**, and the seventh flow rate **F7** represents the flow rate of the liquid that has been discharged from the head **100**, the liquid circulation device of the present embodiment preferably satisfies the relationship of $F5 > (F2 + F6 - F7)$.

[0121] With such a configuration, the internal pressure of the collection tank **220** can be adjusted in accordance with the presence and absence of the liquid discharge from the head **100** within a

certain range in consideration of the fluctuation of the internal pressure of the collection tank **220** due to the amount of the liquid discharged from the head **100**.

Second Embodiment

[0122] FIG. **9** is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a second embodiment of the present disclosure. Arrows in FIG. **9** indicate the direction of flow of liquid or gas.

[0123] The liquid circulation device of the present embodiment includes a variable restrictor **90a** (i.e., a flow rate regulator) that can change the flow rate of the gas flowing through the bypass **90**.

[0124] The variable restrictor **90a** is not limited to any particular restrictor as long as the restrictor has a resistance to the liquid flowing through the bypass **90** and can change the flow rate of the liquid. Preferably, the restrictor **90a** can change the resistance in response to an environmental temperature. In the liquid circulation device of the present embodiment, the flow rate of the gas flowing through the bypass **90** changes in response to the environmental temperature.

[0125] Examples of the liquid discharged from the head **100** include a liquid, such as ink, whose viscosity changes in response to the temperature. In such a liquid, the viscosity decreases and the flow rate of the circulating liquid increases in a high-temperature environment, and the viscosity increases and the flow rate of the circulating liquid decreases in a low-temperature environment.

[0126] For this reason, the flow rate of the gas flowing through the bypass **90** can preferably be adjusted in accordance with the change in the flow rate of the liquid depending on the environmental temperature. For example, the bypass **90** has a cross-sectional area that changes in response to the environmental temperature in a part of the path to change the flow rate of the gas flowing through the bypass **90**.

[0127] Further, the bypass **90** may include an environmental temperature detector to detect the environmental temperature and the flow rate regulator to regulate the flow rate of the gas in response to the environmental temperature detected by the environmental temperature detector.

Third Embodiment

[0128] FIG. **10** is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a third embodiment of the present disclosure. Arrows in FIG. **10** indicate the direction of flow of liquid or gas.

[0129] The liquid circulation device of the present embodiment includes a buffer tank **92** to store an inert gas. The buffer tank **92** communicates with the supply-side air path **83** to supply the inert gas to the gas space **210a** of the supply tank **210** via the supply-side air pump **81**.

[0130] Examples of the inert gas stored in the buffer tank **92** include nitrogen, but are not limited thereto as long as the inert gas can prevent the circulating liquid from changing in quality (deteriorating).

[0131] Such a configuration can prevent the deterioration (degeneration) of the liquid stored in the supply tank **210** due to oxygen or carbon dioxide in the air. Further, the inert gas fed to the collection tank **220** through the bypass **90** can prevent the deterioration of the circulating liquid and the increase in the amount of dissolved oxygen. Accordingly, the liquid can be stably discharged by the head **100**.

Fourth Embodiment

[0132] FIG. **11** is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a fourth embodiment of the present disclosure. Arrows in FIG. **11** indicate the direction of flow of liquid or gas.

[0133] The liquid circulation device of the present embodiment includes the buffer tank **92** to store an inert gas, and the buffer tank **92** communicates with the supply-side air path **83** and the collection-side air path **84**.

[0134] The inert gas supplied to the gas space **210a** of the supply tank **210** via the supply-side air pump **81** is supplied to the collection tank **220** via the bypass **90**, and the inert gas supplied to the collection tank **220** is collected to the buffer tank **92** via the collection-side air pump **82**.

[0135] Similarly to the third embodiment, such a configuration can prevent the deterioration of the circulating liquid and the increase in the amount of dissolved oxygen. As a result, the head **100** can stably discharge the liquid. Further, since the inert gas is circulated via the buffer tank **92**, the amount of use (consumption) of the inert gas can be reduced.

Fifth Embodiment

[0136] FIG. **12** is a schematic diagram illustrating a configuration of a part of a liquid circulation device according to a fifth embodiment of the present disclosure.

[0137] In the liquid circulation device of the present embodiment, the supply-side air pump **81** applies a negative pressure to the supply tank **210**, and the internal pressure of the supply tank **210** is set to a negative pressure lower than that of the collection tank **220**. Specifically, the absolute value of the negative pressure in the supply tank **210** is smaller than the absolute value of the negative pressure in the collection tank **220**.

[0138] The circulation of the liquid via the head **100** is performed by feeding the liquid based on the pressure difference between the supply tank **210** and the collection tank **220**. When the pressure difference applied to the head **100** is small, the internal pressure of the supply tank **210** may be set to a negative pressure as well as the collection tank **220**, as in the present embodiment.

[0139] The head **100** is a circulation-type liquid discharge head. In the circulation-type liquid discharge head, the liquid is supplied from the supply tank **210** to the head **100** through the inlet (i.e., the supply port **171**), and the liquid that has not been discharged from the head **100** is collected to the collection tank **220** through the outlet (i.e., the delivery port **181**).

[0140] For example, the circulation-type liquid discharge head may be an individual liquid chamber type head in which liquid circulates in the vicinity of nozzles or a head in which liquid circulates only in a common liquid chamber. However, when the pressure difference between the supply tank **210** and the collection tank **220** is small as in the present embodiment, a circulation-type liquid discharge head in which liquid circulates only in a common liquid chamber is preferable.

Liquid Circulation Method

[0141] A liquid circulation method of the present embodiment is a method including making the flow rate of the gas flowing through the bypass **90** smaller than the maximum liquid flow rate of the liquid flowing through the liquid feed pump **202** in the liquid circulation device **200**. The liquid circulation device **200** includes the supply tank **210** and the collection tank **220** to store a liquid, the head **100** to discharge the liquid, the path to circulate the liquid via the head **100**, the liquid feed pump **202** disposed in the liquid path **203** to feed the liquid from the collection tank **220** to the supply tank **210**, the supply-side air pump **81** disposed in the supply-side air path **83** communicating with the gas space **210a** of the supply tank **210**, the collection-side air pump **82** disposed in the collection-side air path **84** communicating with the gas space **220a** of the collection tank **220**, and the bypass **90** connecting the gas space **210a** of the supply tank **210** and the gas space **220a** of the collection tank **220** and serving as the flow path for the gas flowing from the supply tank **210** to the collection tank **220**.

[0142] The liquid circulation method of the present embodiment further includes controlling the maximum flow rate of the gas flowing through the supply-side air pump **81** to be larger than the flow rate of the gas flowing through the bypass **90**.

[0143] The flow rate of the gas flowing through the supply-side air pump **81** is preferably made larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid supplied from the supply tank **210** to the head **100**.

[0144] The flow rate of the gas flowing through the supply-side air pump **81** is preferably made larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid supplied from the supply tank **210** to the head **100**, which is increased by the amount of the liquid that has been discharged from the head **100**.

[0145] The liquid circulation method of the present embodiment further includes controlling the

maximum flow rate of the gas flowing through the collection-side air pump **82** to be larger than the flow rate of the gas flowing through the bypass **90**.

[0146] The flow rate of the gas flowing through the collection-side air pump **82** is preferably made larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid collected into the collection tank **220** via the head **100**.

[0147] The flow rate of the gas flowing through the collection-side air pump **82** is preferably made larger than the sum of the flow rate of the gas flowing through the bypass **90** and the flow rate of the liquid collected into the collection tank **220** via the head **100**, which is decreased by the amount of the liquid that has been discharged from the head **100**.

[0148] In the liquid circulation method of the present embodiment, the flow rate of the gas flowing through the bypass is preferably controlled in accordance with the environmental temperature.

[0149] The liquid circulation device including a buffer tank that stores an inert gas preferably supplies the inert gas to the gas space **210a** of the supply tank **210** via the supply-side air pump **81**.

[0150] In the present disclosure, the term “liquid discharge apparatus” includes a liquid discharge head and drives the liquid discharge head to discharge liquid, and is not limited to the printer **1000** illustrated in FIG. **1**.

[0151] In the present disclosure, the liquid is not limited to a particular liquid as long as the liquid has a viscosity or surface tension to be discharged from a head (liquid discharge head). However, preferably, the viscosity of the liquid is not greater than 30 millipascal-second (mPa·s) under ordinary temperature and ordinary pressure or by heating or cooling. Examples of the liquid to be discharged include a solution, a suspension, or an emulsion including, for example, a solvent, such as water or an organic solvent; a colorant, such as dye or pigment; a functional material, such as a polymerizable compound, a resin, or a surfactant; a biocompatible material, such as deoxyribonucleic acid (DNA), amino acid, protein, or calcium; and an edible material, such as a natural colorant. Such a solution, a suspension, or an emulsion can be used for, e.g., inkjet ink; surface treatment liquid; a liquid for forming an electronic element component, a light-emitting element component, or an electronic circuit resist pattern; or a material solution for three-dimensional fabrication.

[0152] In the “liquid discharge head,” examples of an energy source for generating energy to discharge liquid include a piezoelectric actuator (a laminated piezoelectric element or a thin-film piezoelectric element), a thermal actuator that employs a thermoelectric transducer element, such as a thermal resistor, and an electrostatic actuator including a diaphragm and opposed electrodes.

[0153] The term “liquid discharge apparatus” used herein represents an apparatus that drives the liquid discharge head to discharge liquid. The term “liquid discharge apparatus” used herein includes, in addition to apparatuses to discharge liquid to a medium onto which liquid can adhere, apparatuses to discharge the liquid into gas (air) or a different liquid.

[0154] For example, the “liquid discharge apparatus” may further include devices relating to feeding, conveying, and ejecting of the medium onto which liquid can adhere and also include a pretreatment device and an aftertreatment device.

[0155] The “liquid discharge apparatus” may be, for example, an image forming apparatus to form an image on a sheet by discharging ink, or a three-dimensional fabrication apparatus to discharge fabrication liquid to a powder layer in which powder material is formed in layers to form a three-dimensional object.

[0156] The “liquid discharge apparatus” is not limited to an apparatus that discharges liquid to visualize meaningful images such as letters or figures. For example, the liquid discharge apparatus may be an apparatus that forms patterns having no meaning or an apparatus that fabricates three-dimensional images.

[0157] The above-described term “medium onto which liquid can adhere” represents a medium on which liquid is at least temporarily adhered, a medium on which liquid is adhered and fixed, or a medium into which liquid adheres and permeates. Specific examples of the “medium onto which

liquid can adhere” include, but are not limited to, a recording medium such as a paper sheet, recording paper, a recording sheet of paper, a film, or cloth, an electronic component such as an electronic substrate or a piezoelectric element, and a medium such as layered powder, an organ model, or a testing cell. The “medium onto which liquid can adhere” includes any medium to which liquid adheres, unless otherwise specified.

[0158] Examples of materials for the “medium onto which liquid can adhere” include any materials to which liquid can adhere even temporarily, such as paper, thread, fiber, fabric, leather, metal, plastic, glass, wood, and ceramic.

[0159] The term “liquid discharge apparatus” may be an apparatus in which the liquid discharge head and the medium onto which liquid can adhere move relative to each other. However, the liquid discharge apparatus is not limited to such an apparatus. For example, the liquid discharge apparatus may be a serial head apparatus that moves the liquid discharge head or a line head apparatus that does not move the liquid discharge head.

[0160] Examples of the liquid discharge apparatus further include: a treatment liquid applying apparatus that discharges a treatment liquid onto a sheet to apply the treatment liquid to the surface of the sheet, for reforming the surface of the sheet; and an injection granulation apparatus that injects a composition liquid, in which a raw material is dispersed in a solution, through a nozzle to granulate fine particle of the raw material.

[0161] The terms “image formation,” “recording,” “printing,” “image printing,” and “fabricating” used herein may be used synonymously with each other.

[0162] Aspects of the present disclosure are, for example, as follows.

Aspect 1

[0163] A liquid circulation device includes a supply tank and a collection tank to store a liquid, a head to discharge the liquid, a path to circulate the liquid via the head, a liquid feed pump disposed in a liquid path to feed the liquid from the collection tank to the supply tank, a supply-side air pump disposed in a supply-side air path communicating with a gas space of the supply tank, a collection-side air pump disposed in a collection-side air path communicating with a gas space of the collection tank, and a bypass connecting the gas space of the supply tank and the gas space of the collection tank and serving as the flow path for gas flowing from the supply tank to the collection tank. The flow rate of the gas flowing through the bypass is smaller than the maximum flow rate of the liquid flowing through the liquid feed pump.

[0164] In other words, a liquid circulation device includes a supply tank to store a liquid to be supplied to a head and a gas in a first space above a surface of the liquid in the supply tank, a collection tank to store the liquid collected from the head and the gas in a second space above a surface of the liquid in the collection tank, a circulation path including a first circulation path connecting the supply tank and an inlet of the head and a second circulation path connecting an outlet of the head and the collection tank to circulate the liquid through the head, a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas path communicating with the first space in the supply tank, a first gas pump to supply gas to the first space through the first gas path, a second gas path communicating with the second space in the collection tank, a second gas pump to vacuum the gas from the collection tank through the second gas path, a gas bypass connecting the first space in the supply tank and the second space in the collection tank, and circuitry to control the liquid feed pump to flow the liquid through the liquid path at a first flow rate and control the first gas pump and the second gas pump to flow the gas from the first space to the second space through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

Aspect 2

[0165] In the liquid circulation device according to Aspect 1, a maximum flow rate of the gas flowing through the supply-side air pump is larger than the flow rate of the gas flowing through the

bypass.

[0166] In other words, the circuitry is further configured to control the first gas pump to supply the gas to the supply tank at a third flow rate, a maximum value of which is larger than the second flow rate.

Aspect 3

[0167] In the liquid circulation device according to Aspect 1 or 2, a flow rate of the gas flowing through the supply-side air pump is larger than a sum of the flow rate of the gas flowing through the bypass and a flow rate of the liquid supplied from the supply tank to the head.

[0168] In other words, the circuitry is further controls the first gas pump to supply the gas to the supply tank at a third flow rate, controls the first gas pump and the liquid feed pump to supply the liquid from the supply tank to the head through the first circulation path at a fourth flow rate, and controls the first gas pump to control the third flow rate to be larger than a sum of the second flow rate and the fourth flow rate.

Aspect 4

[0169] In the liquid circulation device according to any one of Aspects 1 to 3, a flow rate of the gas flowing through the supply-side air pump is larger than a sum of the flow rate of the gas flowing through the bypass and a flow rate of the liquid supplied from the supply tank to the head, which is increased by the amount of the liquid that has been discharged from the head.

[0170] In other words, in the liquid discharge apparatus according to Aspect 14, the circuitry controls the first gas pump to supply the gas to the supply tank at a third flow rate, controls the first gas pump and the liquid feed pump to supply the liquid from the supply tank to the head through the first circulation path at a fourth flow rate, and controls the first gas pump to control the third flow rate to be larger than a sum of the second flow rate and the fourth flow rate. The fourth flow rate is increased by a seventh flow rate of the liquid that has been discharged from the head.

Aspect 5

[0171] In the liquid circulation device according to any one of Aspects 1 to 4, a maximum flow rate of the gas flowing through the collection-side air pump is larger than the flow rate of the gas flowing through the bypass.

[0172] In other words, the circuitry is further configured to control the second gas pump to vacuum the gas from the collection tank at a fifth flow rate, a maximum value of which is larger than the second flow rate.

Aspect 6

[0173] In the liquid circulation device according to any one of Aspects 1 to 5, a flow rate of the gas flowing through the collection-side air pump is larger than a sum of the flow rate of the gas flowing through the bypass and a flow rate of the liquid collected into the collection tank via the head.

[0174] In other words, the circuitry controls the second gas pump to vacuum the gas from the collection tank at a fifth flow rate, controls the second gas pump and the liquid feed pump to collect the liquid from the head to the collection tank through the second circulation path at a sixth flow rate, and controls the second gas pump to control the fifth flow rate to be larger than a sum of the second flow rate and the sixth flow rate.

Aspect 7

[0175] In the liquid circulation device according to any one of Aspects 1 to 6, a flow rate of the gas flowing through the collection-side air pump is larger than a sum of the flow rate of the gas flowing through the bypass and a flow rate of the liquid collected into the collection tank via the head, which is decreased by the amount of the liquid that has been discharged from the head.

[0176] In other words, in the liquid discharge apparatus according to Aspect 14, the circuitry controls the second gas pump to vacuum the gas from the collection tank at a fifth flow rate, controls the second gas pump and the liquid feed pump to collect the liquid from the head to the collection tank through the second circulation path at a sixth flow rate, and controls the second gas pump to control the fifth flow rate to be larger than a sum of the second flow rate and the sixth flow

rate. The sixth flow rate is decreased by a seventh flow rate of the liquid that has been discharged from the head.

Aspect 8

[0177] In the liquid circulation device according to any one of Aspects 1 to 7, the flow rate of the gas flowing through the bypass changes in response to an environmental temperature.

[0178] In other words, the circuitry further configured to change the second flow rate in response to an environmental temperature.

Aspect 9

[0179] In the liquid circulation device according to Aspect 8, the bypass has a flow path whose cross-sectional area changes in response to the environmental temperature.

[0180] In other words, the circuitry changes a cross-sectional area of the gas bypass in response to the environmental temperature.

Aspect 10

[0181] In the liquid circulation device according to Aspect 8, the bypass includes an environmental temperature detector and a unit that controls a flow rate of the gas in accordance with a detection result of the environmental temperature detector.

[0182] In other words, the gas bypass includes an environmental temperature detector to detect the environmental temperature and a flow rate regulator to regulate the second flow rate. The circuitry controls the flow rate regulator to change the second flow rate in response to the environmental temperature detected by the environmental temperature detector.

Aspect 11

[0183] The liquid circulation device according to any one of Aspects 1 to 10, further includes a buffer tank to store an inert gas. The buffer tank communicates with the supply-side air path. The inert gas is supplied to the gas space of the supply tank via the supply-side air pump.

[0184] In other words, the liquid circulation device according to any one of Aspects 1 to 10, further includes a buffer tank to store an inert gas. The buffer tank communicates with the first gas path, and the first gas pump supplies the inert gas to the first space in the supply tank through the first gas path.

Aspect 12

[0185] In the liquid circulation device according to Aspect 11, the buffer tank communicates with the supply-side air path and the collection-side air path. The inert gas supplied to the supply tank is supplied to the collection tank via the bypass. The inert gas supplied to the collection tank is collected into the buffer tank via the collection-side air pump.

[0186] In other words, the buffer tank further communicates with the second gas path. The inert gas supplied to the supply tank is supplied to the collection tank via the gas bypass. The second gas pump collects the inert gas from the collection tank to the buffer tank through the second gas path.

Aspect 13

[0187] In the liquid circulation device according to any one of Aspects 1 to 12, the head is a circulation-type liquid discharge head.

[0188] In other words, in the liquid discharge apparatus according to Aspect 14, the liquid is supplied from the supply tank to the head through the inlet, and the liquid that has not been discharged from the head is collected to the collection tank through the outlet.

Aspect 14

[0189] A liquid discharge apparatus includes the liquid circulation device according to any one of Aspects 1 to 13.

[0190] In other words, a liquid discharge apparatus includes the liquid circulation device according to any one of Aspects 1 to 13 and a head having the inlet connected to the first circulation path and the outlet connected to the second circulation path to discharge the liquid.

Aspect 15

[0191] An image forming apparatus includes the liquid circulation device according to any one of

Aspects 1 to 13.

[0192] In other words, an image forming apparatus includes the liquid circulation device according to any one of Aspects 1 to 13, a head having the inlet connected to the first circulation path and the outlet connected to the second circulation path to discharge the liquid onto a medium, and a conveyor to convey the medium to the head.

Aspect 16

[0193] A liquid discharge apparatus includes a head to discharge a liquid, a supply tank to store the liquid and a gas, a collection tank to store the liquid and the gas, and a circulation path to circulate the liquid. The circulation path includes a first circulation path connecting the supply tank and the head and a second circulation path connecting the head and the collection tank. The liquid discharge head further includes a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas pump to supply gas to the supply tank, a second gas pump to vacuum the gas from the collection tank, and a gas bypass connecting the supply tank and the collection tank.

Aspect 17

[0194] In the liquid discharge apparatus according to Aspect 16, further includes circuitry that controls the liquid feed pump to flow the liquid through the liquid path at a first flow rate and controls the first gas pump and the second gas pump to flow the gas from the supply tank to the collection tank through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

[0195] Aspect 18 In the liquid discharge apparatus according to Aspect 16 or 17, the gas in the supply tank is in a first space above a surface of the liquid in the supply tank, and the gas in the collection tank is in a second space above a surface of the liquid in the collection tank. The gas bypass connects the first space and the second space.

Aspect 19

[0196] An image forming apparatus includes a head to discharge a liquid onto a medium, a supply tank to store the liquid and a gas, a collection tank to store the liquid and the gas, and a circulation path to circulate the liquid. The circulation path includes a first circulation path connecting the supply tank and the head and a second circulation path connecting the head and the collection tank. The image forming apparatus further include a liquid path connecting the supply tank and the collection tank, a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path, a first gas pump to supply gas to the supply tank, a second gas pump to vacuum the gas from the collection tank, a gas bypass connecting the supply tank and the collection tank, and circuitry that controls the liquid feed pump to flow the liquid through the liquid path at a first flow rate and controls the first gas pump and the second gas pump to flow the gas from the supply tank to the collection tank through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

Aspect 20

[0197] The image forming apparatus according to Aspect 19, further comprising a conveyor to convey the medium.

Aspect 21

[0198] A liquid circulation method includes making a flow rate of a gas flowing through a bypass smaller than a maximum liquid flow rate of a liquid flowing through a liquid feed pump in the liquid circulation device. The liquid circulation device includes a supply tank and a collection tank to store the liquid, a head to discharge the liquid, a path to circulate the liquid via the head, the liquid feed pump disposed in a liquid path to feed the liquid from the collection tank to the supply tank, a supply-side air pump disposed in a supply-side air path communicating with a gas space of the supply tank, a collection-side air pump disposed in a collection-side air path communicating with a gas space of the collection tank, and the bypass connecting the gas space of the supply tank and the gas space of the collection tank and serving as the flow path for the gas flowing from the

supply tank to the collection tank.

[0199] As described above, according to one aspect of the present disclosure, a liquid circulation device can be provided that reduces, with a simple configuration, the pulsation of pressure that is generated when a pump is driven to feed the liquid from the collection tank to the supply tank.

[0200] The above-described embodiments are illustrative and do not limit the present invention.

Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention.

[0201] Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

[0202] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, ASICs (“Application Specific Integrated Circuits”), FPGAs (“Field-Programmable Gate Arrays”), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.

[0203] There is a memory that stores a computer program which includes computer instructions. These computer instructions provide the logic and routines that enable the hardware (e.g., processing circuitry or circuitry) to perform the method disclosed herein. This computer program can be implemented in known formats as a computer-readable storage medium, a computer program product, a memory device, a record medium such as a CD-ROM or DVD, and/or the memory of a FPGA or ASIC.

Claims

1. A liquid circulation device comprising: a supply tank to store a liquid to be supplied to a head and a gas in a first space above a surface of the liquid in the supply tank; a collection tank to store the liquid collected from the head and the gas in a second space above a surface of the liquid in the collection tank; a circulation path to circulate the liquid through the head, the circulation path including: a first circulation path connecting the supply tank and an inlet of the head; and a second circulation path connecting an outlet of the head and the collection tank; a liquid path connecting the supply tank and the collection tank; a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path; a first gas path communicating with the first space in the supply tank; a first gas pump to supply gas to the first space through the first gas path; a second gas path communicating with the second space in the collection tank; a second gas pump to vacuum the gas from the collection tank through the second gas path; a gas bypass connecting the first space in the supply tank and the second space in the collection tank; and circuitry configured to: control the liquid feed pump to flow the liquid through the liquid path at a first flow rate; and control the first gas pump and the second gas pump to flow the gas from the first space to the second space through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

2. The liquid circulation device according to claim 1, wherein the circuitry is further configured to control the first gas pump to supply the gas to the supply tank at a third flow rate, a maximum value of which is larger than the second flow rate.

3. The liquid circulation device according to claim 1, wherein the circuitry is further configured to: control the first gas pump to supply the gas to the supply tank at a third flow rate; control the first gas pump and the liquid feed pump to supply the liquid from the supply tank to the head through

the first circulation path at a fourth flow rate; and control the first gas pump to control the third flow rate to be larger than a sum of the second flow rate and the fourth flow rate.

4. The liquid circulation device according to claim 1, wherein the circuitry is further configured to control the second gas pump to vacuum the gas from the collection tank at a fifth flow rate, a maximum value of which is larger than the second flow rate.
5. The liquid circulation device according to claim 1, wherein the circuitry is further configured to: control the second gas pump to vacuum the gas from the collection tank at a fifth flow rate; control the second gas pump and the liquid feed pump to collect the liquid from the head to the collection tank through the second circulation path at a sixth flow rate; and control the second gas pump to control the fifth flow rate to be larger than a sum of the second flow rate and the sixth flow rate.
6. The liquid circulation device according to claim 1, wherein the circuitry further configured to change the second flow rate in response to an environmental temperature.
7. The liquid circulation device according to claim 6, wherein the circuitry further configured to change a cross-sectional area of the gas bypass in response to the environmental temperature.
8. The liquid circulation device according to claim 6, wherein the gas bypass includes: an environmental temperature detector to detect the environmental temperature; and a flow rate regulator to regulate the second flow rate; and the circuitry further configured to control the flow rate regulator to change the second flow rate in response to the environmental temperature detected by the environmental temperature detector.
9. The liquid circulation device according to claim 1, further comprising a buffer tank to store an inert gas, wherein the buffer tank communicates with the first gas path, and the first gas pump supplies the inert gas to the first space in the supply tank through the first gas path.
10. The liquid circulation device according to claim 9, wherein the buffer tank further communicates with the second gas path, the inert gas supplied to the supply tank is supplied to the collection tank via the gas bypass, and the second gas pump collects the inert gas from the collection tank to the buffer tank through the second gas path.
11. A liquid discharge apparatus comprising: the liquid circulation device according to claim 1; and a head having the inlet connected to the first circulation path and the outlet connected to the second circulation path to discharge the liquid.
12. The liquid discharge apparatus according to claim 11, wherein the liquid is supplied from the supply tank to the head through the inlet, and the liquid that has not been discharged from the head is collected to the collection tank through the outlet.
13. The liquid discharge apparatus according to claim 11, wherein the circuitry is further configured to: control the first gas pump to supply the gas to the supply tank at a third flow rate; control the first gas pump and the liquid feed pump to supply the liquid from the supply tank to the head through the first circulation path at a fourth flow rate; and control the first gas pump to control the third flow rate to be larger than a sum of the second flow rate and the fourth flow rate, the fourth flow rate increased by a seventh flow rate of the liquid that has been discharged from the head.
14. The liquid discharge apparatus according to claim 11, wherein the circuitry is further configured to: control the second gas pump to vacuum the gas from the collection tank at a fifth flow rate; control the second gas pump and the liquid feed pump to collect the liquid from the head to the collection tank through the second circulation path at a sixth flow rate; and control the second gas pump to control the fifth flow rate to be larger than a sum of the second flow rate and the sixth flow rate, the sixth flow rate decreased by a seventh flow rate of the liquid that has been discharged from the head.
15. An image forming apparatus comprising: the liquid circulation device according to claim 1; a head having the inlet connected to the first circulation path and the outlet connected to the second circulation path to discharge the liquid onto a medium; and a conveyor to convey the medium to the head.

16. A liquid discharge apparatus comprising: a head to discharge a liquid; a supply tank to store the liquid and a gas; a collection tank to store the liquid and the gas; a circulation path to circulate the liquid, the circulation path including: a first circulation path connecting the supply tank and the head; and a second circulation path connecting the head and the collection tank; a liquid path connecting the supply tank and the collection tank; a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path; a first gas pump to supply gas to the supply tank; a second gas pump to vacuum the gas from the collection tank; and a gas bypass connecting the supply tank and the collection tank.

17. The liquid discharge apparatus according to claim 16, further comprising circuitry configured to: control the liquid feed pump to flow the liquid through the liquid path at a first flow rate; and control the first gas pump and the second gas pump to flow the gas from the supply tank to the collection tank through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

18. The liquid discharge apparatus according to claim 16, wherein the gas in the supply tank is in a first space above a surface of the liquid in the supply tank, the gas in the collection tank is in a second space above a surface of the liquid in the collection tank, and the gas bypass connecting the first space and the second space.

19. An image forming apparatus comprising: a head to discharge a liquid onto a medium; a supply tank to store the liquid and a gas; a collection tank to store the liquid and the gas; a circulation path to circulate the liquid, the circulation path including: a first circulation path connecting the supply tank and the head; and a second circulation path connecting the head and the collection tank; a liquid path connecting the supply tank and the collection tank; a liquid feed pump to feed the liquid from the collection tank to the supply tank through the liquid path; a first gas pump to supply gas to the supply tank; a second gas pump to vacuum the gas from the collection tank; a gas bypass connecting the supply tank and the collection tank; and circuitry configured to: control the liquid feed pump to flow the liquid through the liquid path at a first flow rate; and control the first gas pump and the second gas pump to flow the gas from the supply tank to the collection tank through the gas bypass at a second flow rate smaller than a maximum value of the first flow rate.

20. The image forming apparatus according to claim 19, further comprising a conveyor to convey the medium.
